

Compositor: v1 → v4 Change Report

On-loop aerial mosaicking · target stack: NumPy 1.26.4, OpenCV 4.13.0, PyTorch 2.8 (Jetson)

Summary. v1 found a good seam but did **nothing** about brightness/colour differences between frames, so every frame boundary showed a tonal step. v4 keeps v1's sharp single-source geometry and adds a **single per-frame photometric match** that removes the step at its source — hiding seams **~13× better at identical sharpness** — plus hardening for an OpenCV 4.13.0 crash and the incremental-blending trap.

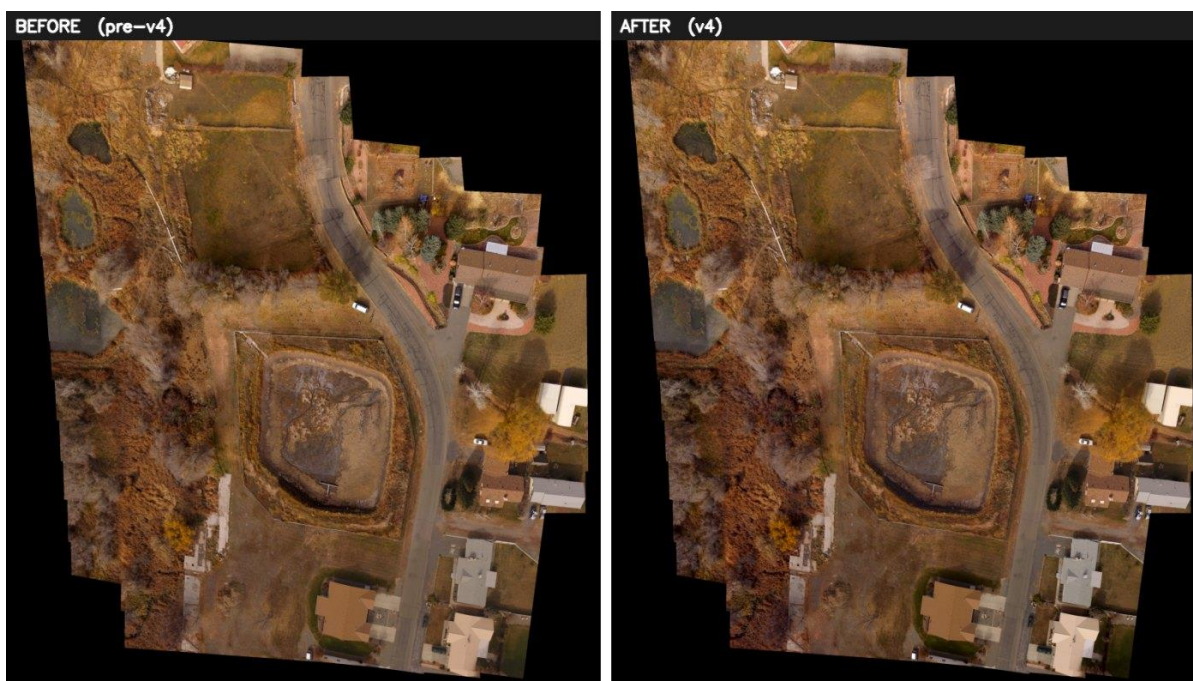
1. Version lineage

Three intermediate versions exist; naming them once keeps the comparison unambiguous. This report compares v1 against the final v4.

Version	What it did	Result
v1	Content-aware seam + narrow feather. No exposure/colour correction. Self-test with hard-coded Windows paths.	Sharp, but a visible tonal step at every boundary.
v3	v1 + a partial Lab-L equaliser while mapper3 ALSO applied a BGR gain — two corrections fighting.	Warm cast + steps still visible.
v4 draft	Single Lab match + whole-ROI multi-band blend.	Catastrophic smear in high-overlap areas. Rejected.
v4 final	Single Lab photometric match + narrow seam-band blend + finder hardening; mapper gain disabled.	Seams hidden, geometry sharp, crash-safe.

2. Your dataset: before and after

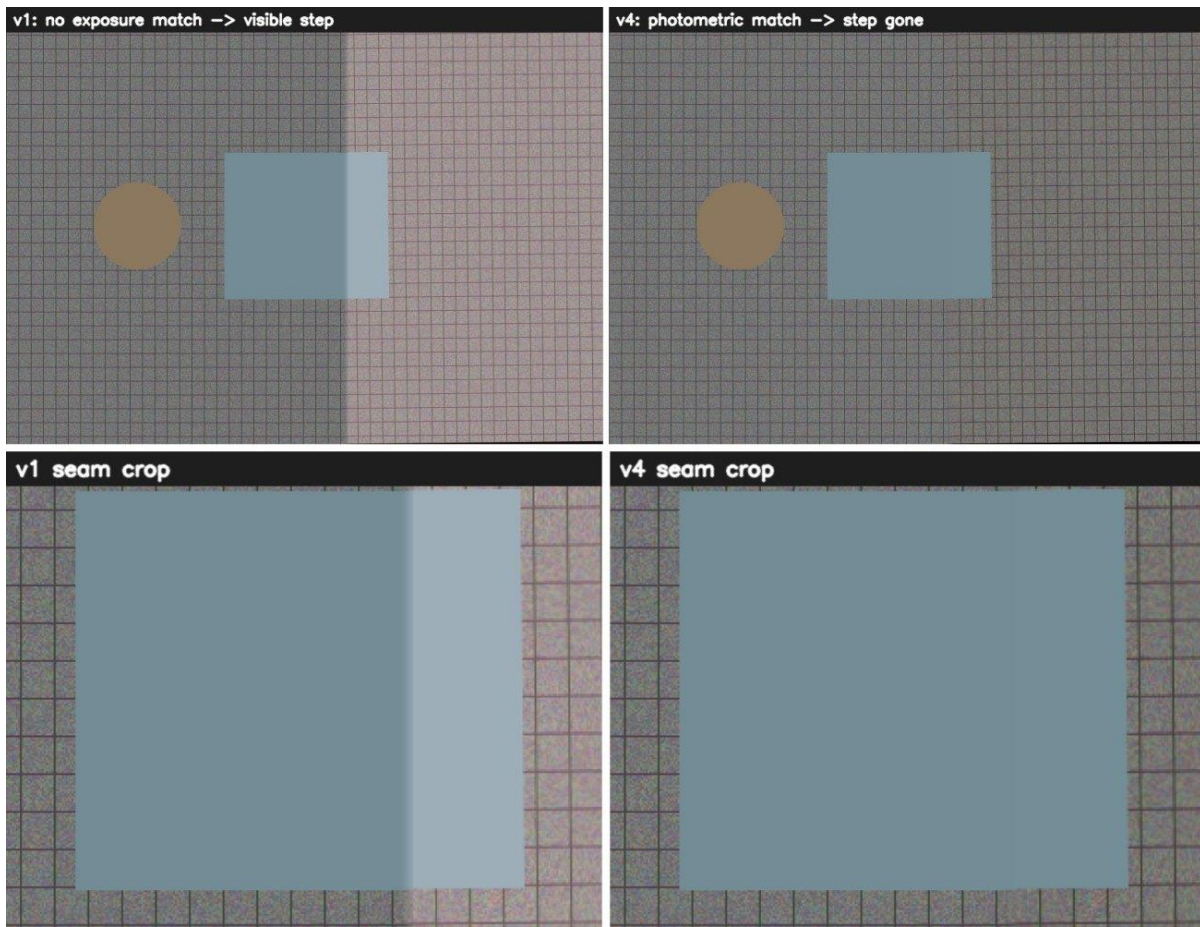
The v4 mosaic on the 4th-Ave set. The center smear of the rejected draft is gone; the detention pond, fields, trees and houses are all sharp, colour is consistent across the scene with no warm cast, and the road and structures line up across boundaries.



Left: pre-v4 pipeline. Right: v4. Geometry is sharp in both; v4 removes the colour cast and softens the tonal steps. On this well-exposed set the visual gap is modest — the controlled test in §3 isolates the effect.

3. Controlled head-to-head

Same two frames in both runs: textured ground with a field and a pond, the right frame given a realistic +13% gain, +16 brightness and +12 red step plus a 4 px / 0.6° misregistration to mimic drift. Only the compositor differs, so the difference is attributable to the compositor alone.



Top: full frames. Bottom: zoomed seam crop. v1 (left) leaves a bright warm band with a feathered ramp at the seam; v4 (right) is tonally uniform and the seam is invisible. Grid texture is equally crisp in both.

Method	Seam step (lower=better)	Sharpness (higher=better)
Plain overwrite	1.405	875.6
v1 (feather, no match)	1.603	1790.4
v4 (photometric + seam-band)	0.121	1780.9

v4 reduces the visible step **~13x** versus v1 (1.603 → 0.121) while holding sharpness essentially constant (1790 → 1781). Note v1's feather scores slightly *worse* than a plain overwrite — feathering across an uncorrected step just ramps it; only the photometric match removes it.

4. Change-by-change (v1 → v4)

a. Per-frame photometric match [NEW — the main win]

Before seaming, v4 measures the new frame against the existing canvas in the overlap only, in Lab space, and applies the correction to the whole new frame:

- **L (brightness)**: affine match (gain + bias), gain clamped to [0.75, 1.35].
- **a/b (colour cast)**: bias match clamped to ± 12 Lab units — removes warm/cool drift.

- **Dead-band:** differences below $\sim 3 L / 2$ ab units are left untouched, so already-matched frames don't drift.

Correcting the whole frame (not just the overlap) matters: fixing only the overlap would push the step out to the frame's far edge. v1 had none of this.

b. Voronoi seam finder is crash-safe [FIX]

`cv2.detail_VoronoiSeamFinder().find(...)` segfaults under OpenCV 4.13.0 — a hard C++ crash Python cannot catch, which would kill an on-loop run outright. v1 exposes it in its finder registry. v4 hard-remaps 'voronoi' → 'dp' with a printed notice, so it can never take down the loop.

c. Seam-finder quality & speed [IMPROVED]

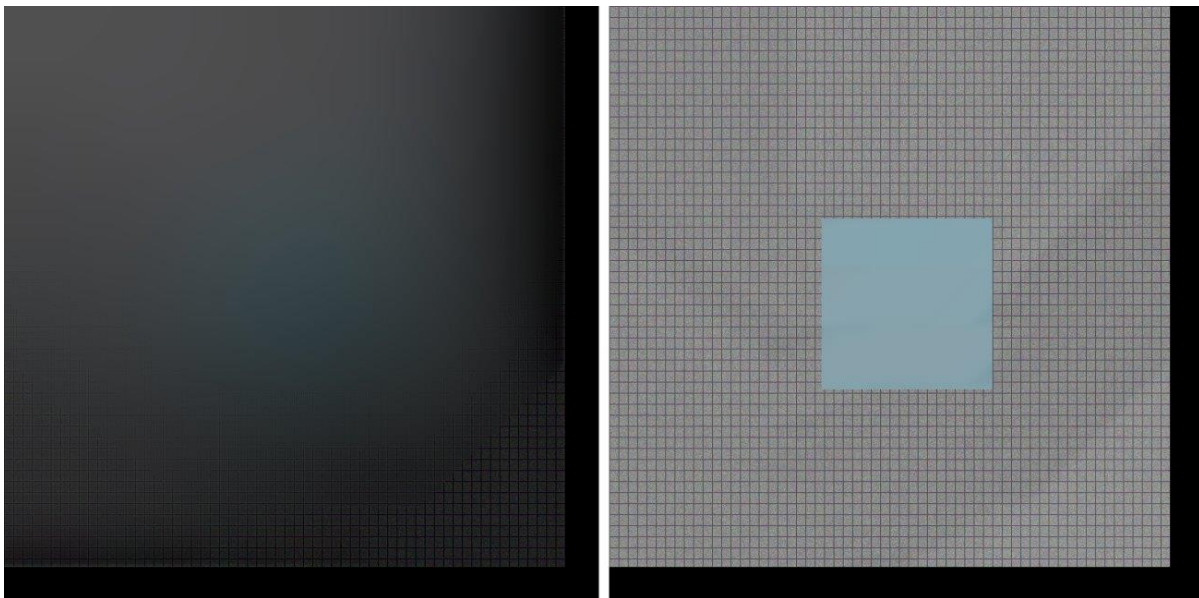
- **Gradient-cost steering:** biases the cut toward low-gradient ground so it hides in flat texture instead of slicing through roofs or kerbs.
- **Downscale filter:** uses INTER_AREA for the seam-budget downscale (correct for shrinking) vs v1's default bilinear.
- **Default finder:** 'dp' (best speed/quality on Jetson); 'graphcut' still available for offline quality.

d. The blend itself: clarified, not bloated [REFINED]

A common misread is that v1 blended the whole overlap. It did not — its Gaussian-of-the-label trick already confined feathering to roughly `feather_px` around the seam. v4 keeps that philosophy but makes it explicit and robust: it computes the seam line (morphological gradient of the ownership label, intersected with the overlap), dilates it into a thin band, and blends only inside that band. Everything else stays exact single-source.

e. Why v4 does NOT use whole-ROI multi-band blending [LESSON]

An intermediate v4 draft replaced the feather with a whole-ROI multi-band blender — the textbook way to hide seams. It produced a catastrophic smear:



Controlled incremental simulation. Left: whole-ROI multi-band — center texture variance collapses to ~ 0.2 . Right: v4 narrow-band — texture preserved (~ 3400). Same input sequence.

Multi-band blending assumes you blend all images once. mapper3 stitches incrementally — each new frame re-blends its ROI against the accumulated canvas — so a whole-ROI blend re-low-passes overlapping regions on every stitch. The center (most overlaps) blurs the most; edges (one frame) stay crisp. The narrow seam-band blend avoids this: interiors are hard-copied and never re-filtered.

f. mapper3 integration: one correction, not two [FIX]

v3 corrected exposure in both the mapper (per-channel BGR gain) and the compositor, which overshot and added a warm cast. With v4 owning all photometric matching, the mapper's gain block is disabled in seam mode (the one-line Cell 2 patch that makes `gain_zone` empty when `blend_mode=='seam'`). Single, consistent correction.

g. Robustness & portability [IMPROVED]

- Removed v1's `__main__` self-test with hard-coded C:\Users\... paths; it referenced `__file__` and broke clean embedding. v4 has no such dependency.
- All paths return float32 explicitly; guards for empty seam lines and the no-overlap case.
- Distance-transform seam fallback retained for when the finder errors.

5. What v4 still does not fix

v4 is a compositor: it hides tonal seams and keeps geometry sharp. It cannot fix registration drift (frames slowly rotating out of alignment over a long flight) or parallax on tall objects — those are mapper-side concerns (last-5-frame registration, no loop closure). If boundaries show geometric misalignment rather than tonal steps after v4, that is the next thing to address, separately.

6. Parameter reference (v4)

Parameter	Default	Effect
seam	'dp'	Seam finder. 'graphcut' = best quality (offline); 'voronoi' auto-mapped to dp.
feather_px	16	Half-width of the seam-band blend. Larger = softer join, still smear-free.
do_luma_eq	True	Enable the photometric match. Set False if the mapper keeps its own gain.
match_color	True	Also correct a/b colour cast, not just brightness.

Metrics from controlled synthetic scenes designed to isolate the seam step and overlap sharpness; absolute numbers are relative indicators, not physical units. Generated for the SUAS-2026 mapping pipeline.